MTH109

Examination - July Semester 2025

Calculus

Tuesday, 18 November 2025 **01:00 pm - 03:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FIVE (5)** questions and comprises **FOUR (4)** printed pages (including cover page).

2. You must answer **ALL** questions.

3. All answers must be written in the answer book. Marks will only be awarded if FULL working is shown.

4. This is an **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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(Full marks: 100)

Question 1

Determine the following limits.

Question 1a

$$\lim_{x \to 2} \frac{24x^3 - 3x^2 + 4}{5 - 3x}$$

(5 marks)

Question 1b

$$\lim_{x \to 24} \frac{5 - \sqrt{x+1}}{x - 24}$$

(6 marks)

Question 1c

$$\lim_{x \to 0} \left(1 + 24x^2\right)^{\frac{3}{x^2}}$$

(7 marks)

Question 1d

$$\lim_{x \to 0} x^{24} \ln\left(1 + \sin^2 \frac{1}{x^{24}}\right)$$

(7 marks)

Question 2a

A certain chemical reaction produces a quantity of product Q (in grams) at time t (in hours).

$$Q(t) = 24te^{-0.5t}, \quad t \geq 0$$

Show clearly that the stationary points of $Q(t)$ occur when t satisfies the equation

$$(2-t)e^{-0.5t} = 0$$

Hence, apply derivative tests to classify the stationary point(s) and find exactly the corresponding maximum/minimum quantity of product.

(15 marks)

Question 2b

Define a function $F : [0, \infty) \longrightarrow \mathbb{R}$ by

$$F(x) = e^{2x} + \ln(1+x).$$

Use an appropriate Taylor polynomial of F to estimate $F(1.01)$ correct to 1 decimal place.

(5 marks)

Question 3a

Solve and simplify the following:

$$\frac{d}{dx}\int_{24}^{x} \sin(\cos t^2)\, dt.$$

(5 marks)

Question 3b

Solve and simplify the following:

$$\frac{d}{dx}\int_{x}^{\sin x} \cos(e^t)\, dt.$$

(15 marks)

Question 4

Employ various techniques of integration to evaluate the following integrals.

Question 4a

$$\int \frac{6x}{(x-2)(x^2+4)}\,dx$$

(7 marks)

Question 4b

$$\int \frac{x^2}{\sqrt{16-x^2}}\,dx$$

(6 marks)

Question 4c

$$\int_0^2 xe^{24x}\,dx$$

(7 marks)

Question 5a

Calculate the area between the curves

$$y = x\sqrt{x^2+1}, y = e^{-x}, x = -4$$

and the y-axis.

(7 marks)

Question 5b

Calculate the volume of the solid obtained by revolving the region between the graphs of $x = y^2 - 6y + 8$ and $x = 3$ about the y-axis.

(8 marks)

----- END OF PAPER -----